

3D Ising Model Simulation

Metropolis Monte Carlo

The 3D Ising model is an extension of the well-known 2D Ising model to three dimensions. Just like in the 2D case, it studies how magnetic materials behave at different temperatures. In this model, a three-dimensional grid of atoms (called spins) can either point up or down, and neighbouring spins interact with each other.

We simulated this model on a 30x30x30 grid using the Metropolis Monte Carlo method, which randomly flips spins and accepts or rejects the flip based on energy rules. The simulation was run at six different temperatures — $T = 1.0, 3.0, 4.0, 4.511, 5.0,$ and 6.0 — to observe how the system behaves when it is cold (ordered), near the critical temperature ($T_c = 4.511$), and hot (disordered). The critical temperature is higher in 3D than in 2D because each spin has more neighbours, making the ordered phase more stable.

At each temperature, 50 independent runs were performed and the following quantities were measured:

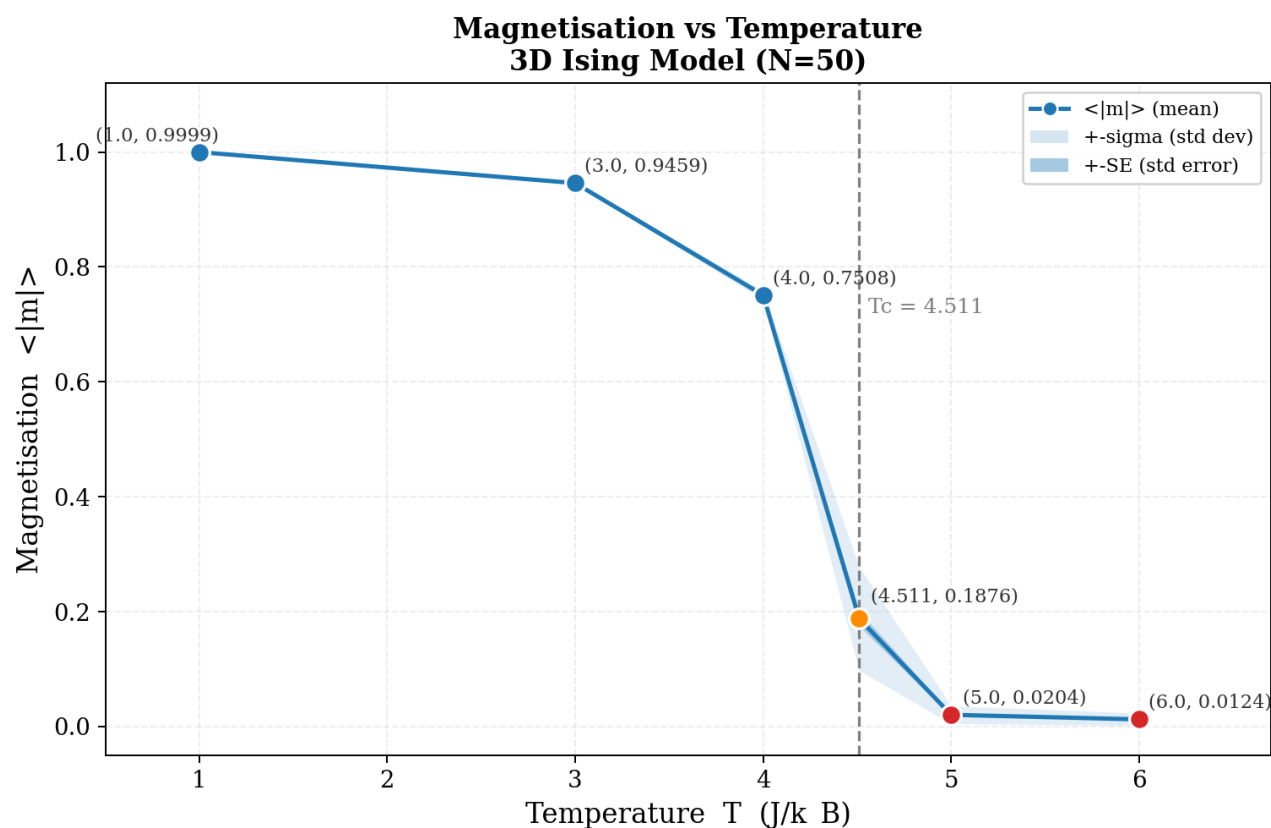
- **Magnetisation** — how aligned the spins are on average
- **Spatial Correlation $G(r)$** — how much two spins influence each other over distance
- **Time Correlation $C(t)$** — how much the system remembers its past state over time
- **Structure Factor $S(k)$** — the spin arrangement pattern in frequency space
- **Cluster Statistics** — how spins group together into connected regions

The shaded bands in each plot show the uncertainty in the results — the wider band is the standard deviation ($\pm\sigma$) and the narrower band is the standard error ($\pm SE$).

Simulation Parameters

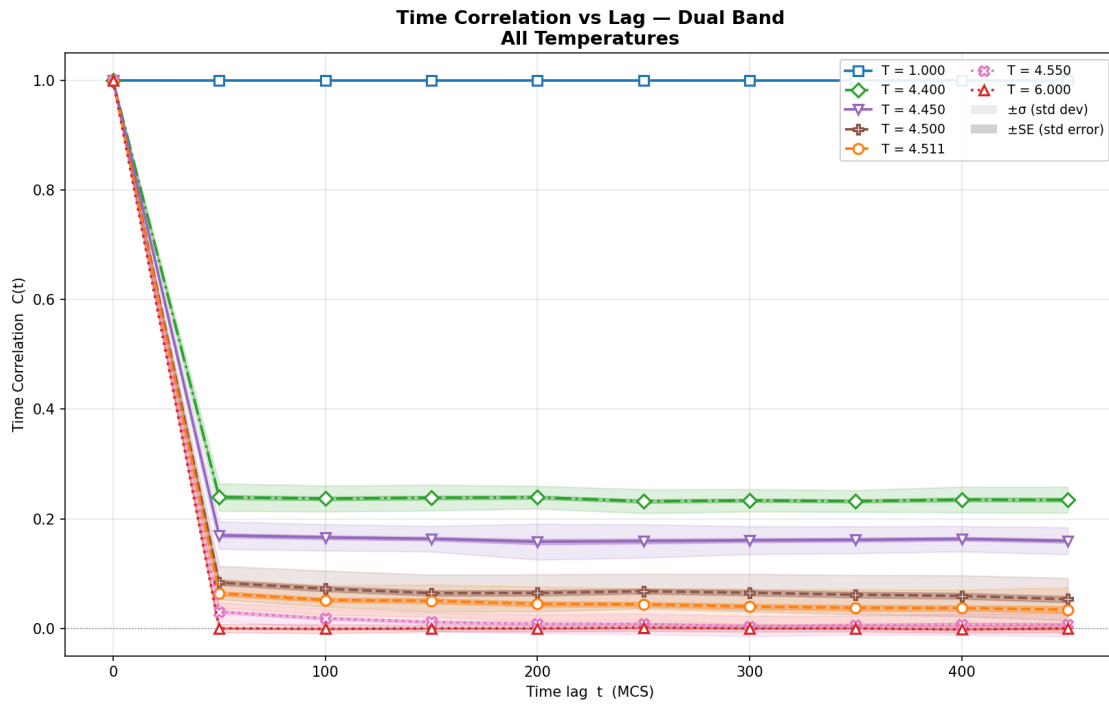
Parameter	Value
Lattice Size	30 x 30 x 30
Algorithm	Metropolis Monte Carlo
Boundary Condition	Periodic Boundary Condition
Equilibration Steps	100,000 MC steps
Number of Samples	50
Measurement Gap	1,000 MC steps
Temperatures Studied	1.0, 3.0, 4.0, 4.511, 5.0, 6.0
Critical Temperature	$T_c = 4.511$

Figure 1 — Magnetisation vs Temperature



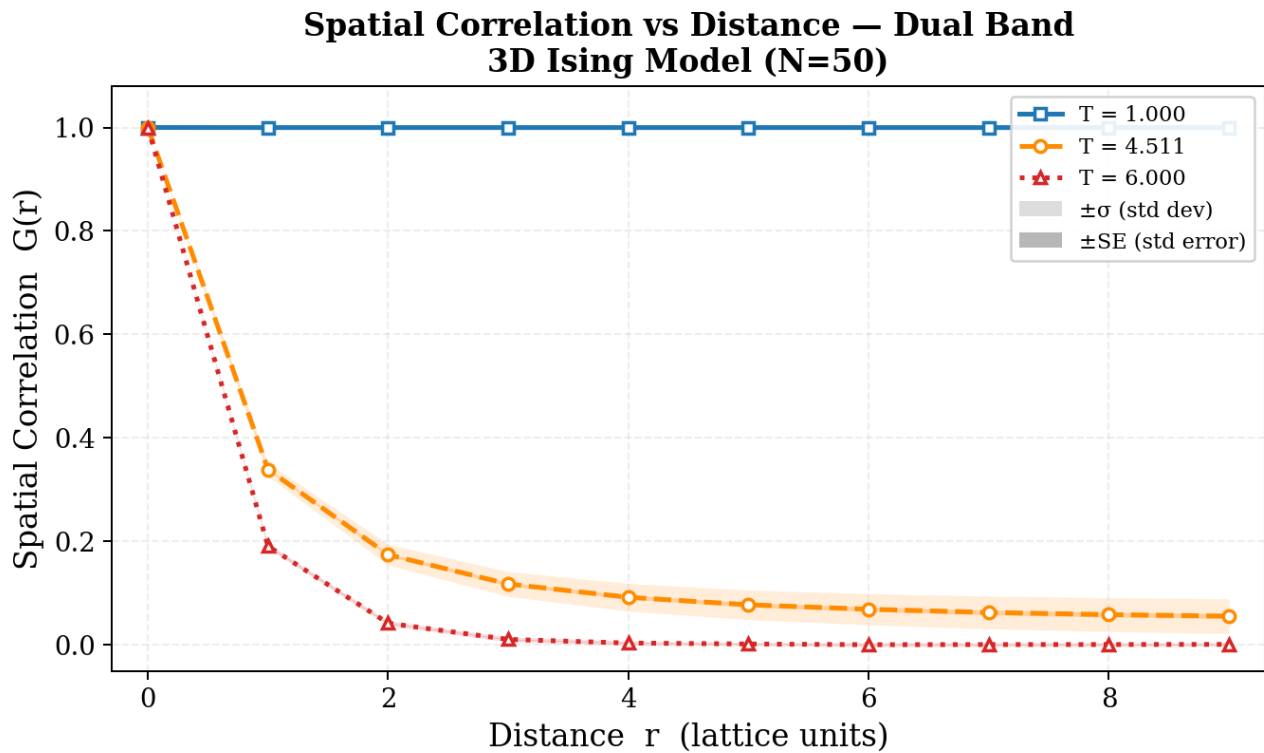
This figure shows the average magnetisation of the system at different temperatures. At $T = 1.0$, almost all spins are aligned, giving magnetisation close to 1. As the temperature increases, the magnetisation decreases steadily. Near the critical temperature ($T_c = 4.511$), the magnetisation falls very rapidly, showing the phase transition. At $T = 5.0$ and $T = 6.0$, the spins become almost random and the magnetisation becomes very small. The shaded region represents the standard deviation and the narrower band shows the standard error.

Figure 2 — Time Correlation Function $C(t)$



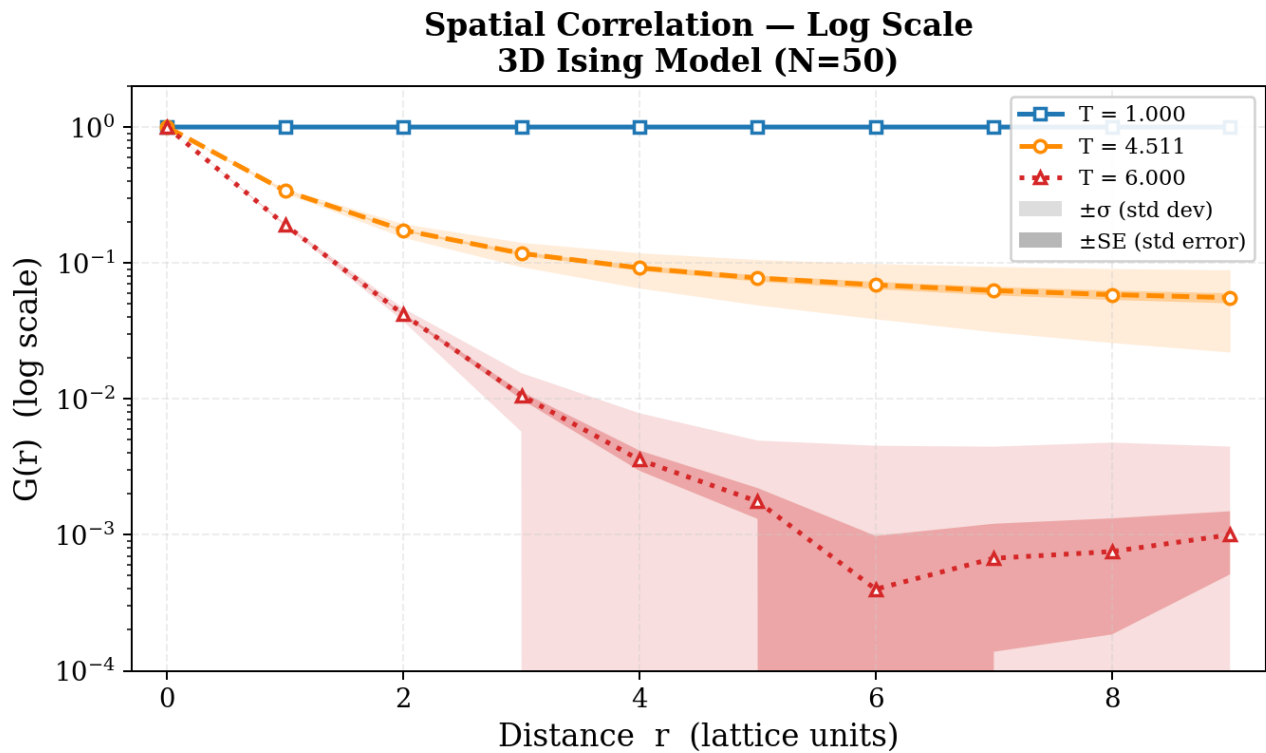
Time correlation function $C(t)$ for the 3D Ising model at seven temperatures spanning the ordered, critical, and disordered phases. At low temperature ($T = 1.0$), the correlation remains close to unity, indicating a stable ordered state. As the temperature approaches the critical point ($T = 4.4$ – 4.511), the correlation decays more rapidly and exhibits the largest fluctuations, reflecting critical slowing down. Above the critical temperature ($T = 4.55$ and 6.0), the correlation quickly approaches zero due to rapid loss of spin order.

Figure 3 — Spatial Correlation Function $G(r)$



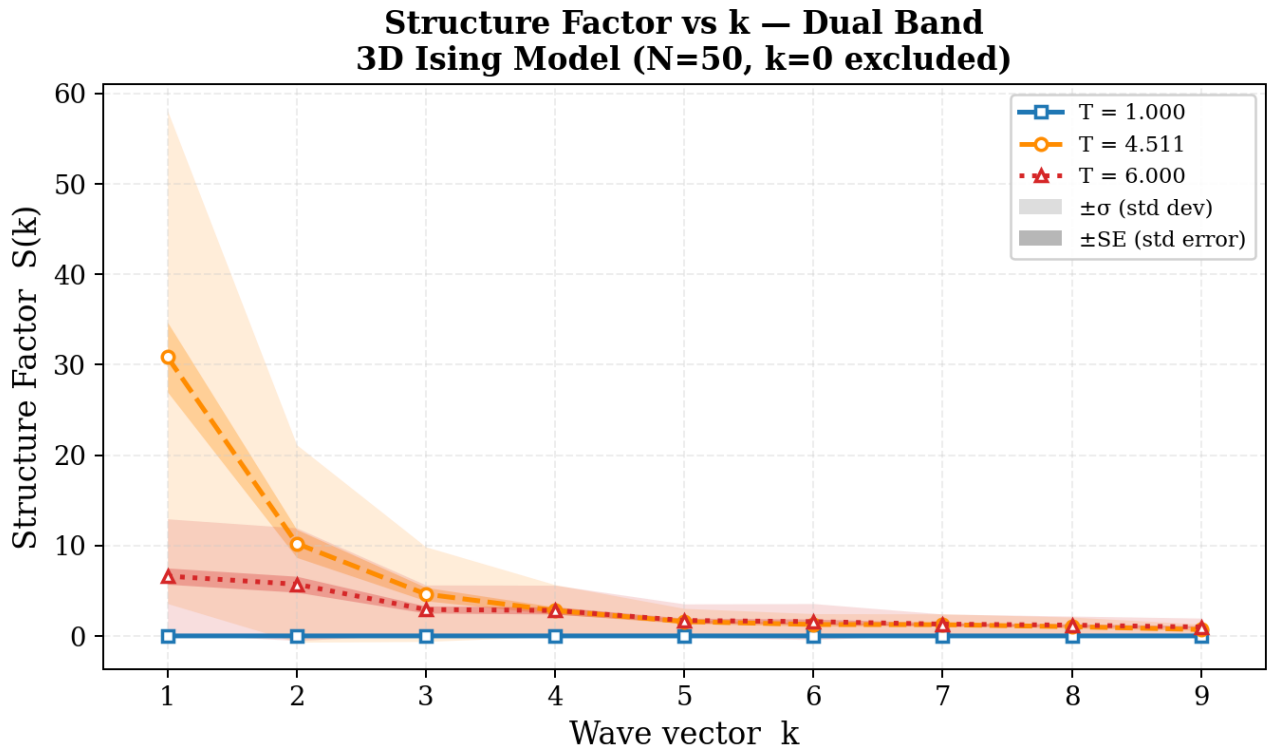
This figure shows how strongly two spins are related when they are separated by a distance r . At $T = 1.0$, the correlation remains close to 1 even at large distances because the entire lattice is ordered. At $T_c = 4.511$, the correlation decreases slowly with distance, indicating the presence of large domains. At $T = 6.0$, the correlation decreases very rapidly, showing that distant spins become almost independent. The increase in uncertainty at larger distances is shown by the standard deviation and error bands.

Figure 4 — Spatial Correlation (Log Scale)



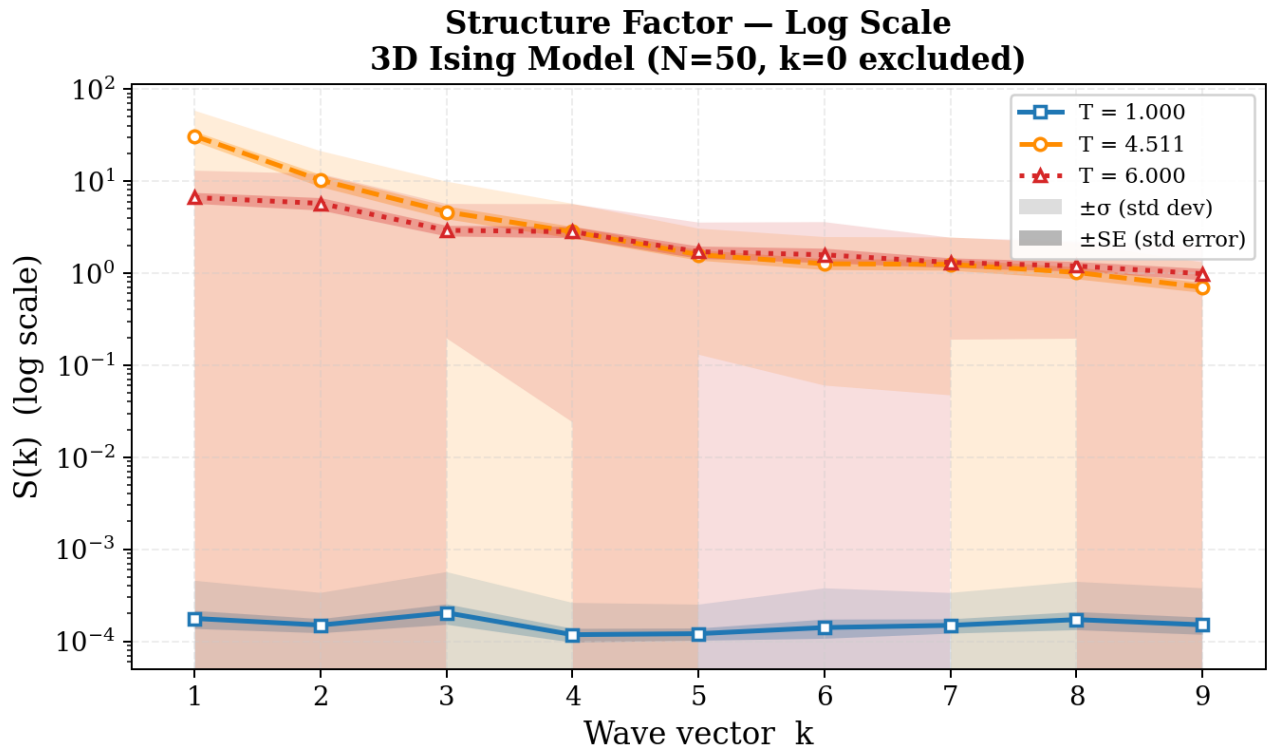
The logarithmic scale makes the small values of the correlation function visible. The low-temperature data remain nearly constant because the system is ordered. Near the critical temperature, the correlation decreases gradually, while above the critical temperature it decreases very rapidly. The larger spread in the error bands at large distances occurs because the correlation values become very small and fluctuate more strongly.

Figure 5 — Structure Factor $S(k)$



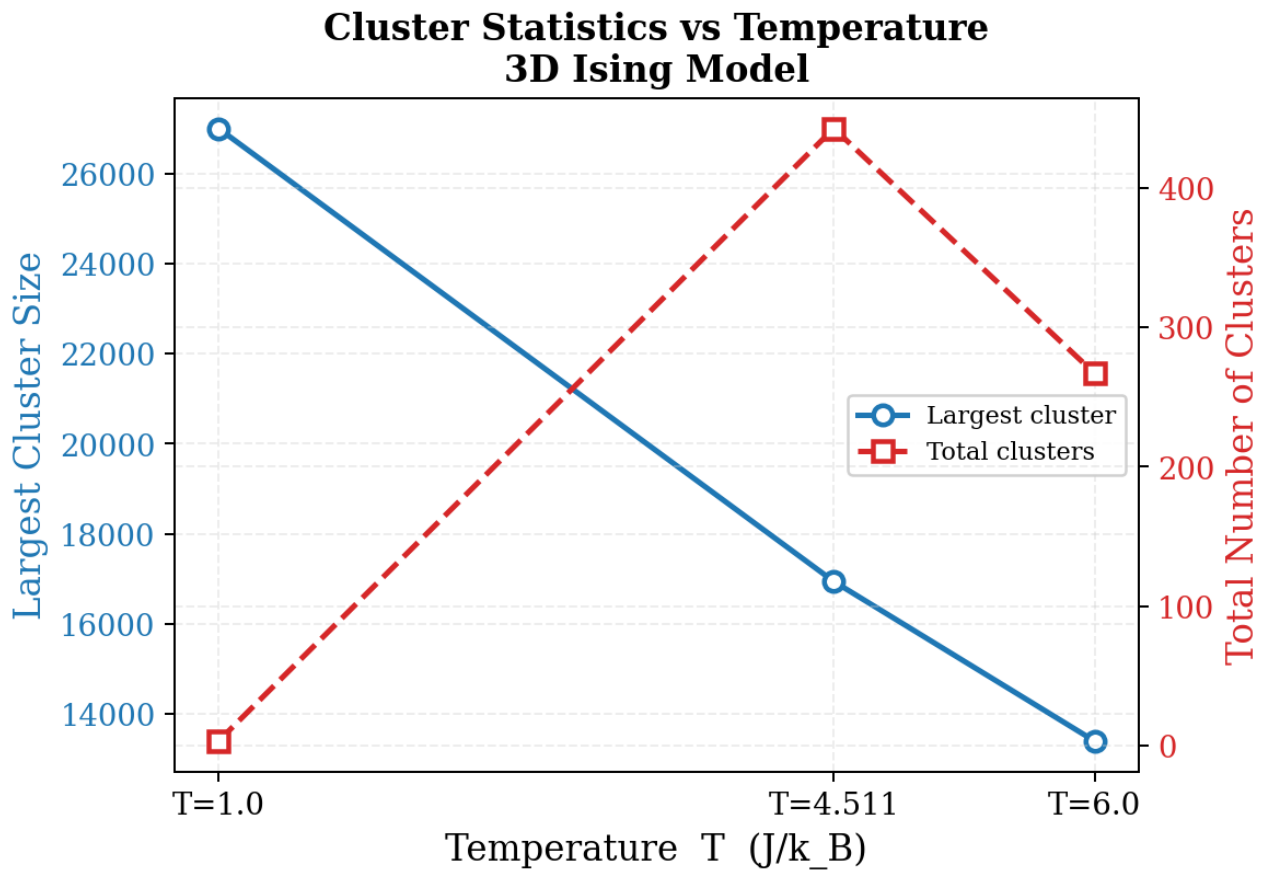
The structure factor gives information about the arrangement of spins at different length scales. The low-temperature state shows very small values because the system is highly ordered. Near the critical temperature, the structure factor becomes large at small values of k , indicating the formation of large spin domains. At high temperature the values become smaller because the system becomes disordered. The fluctuations are largest near the critical temperature.

Figure 6 — Structure Factor $S(k)$ (Log Scale)



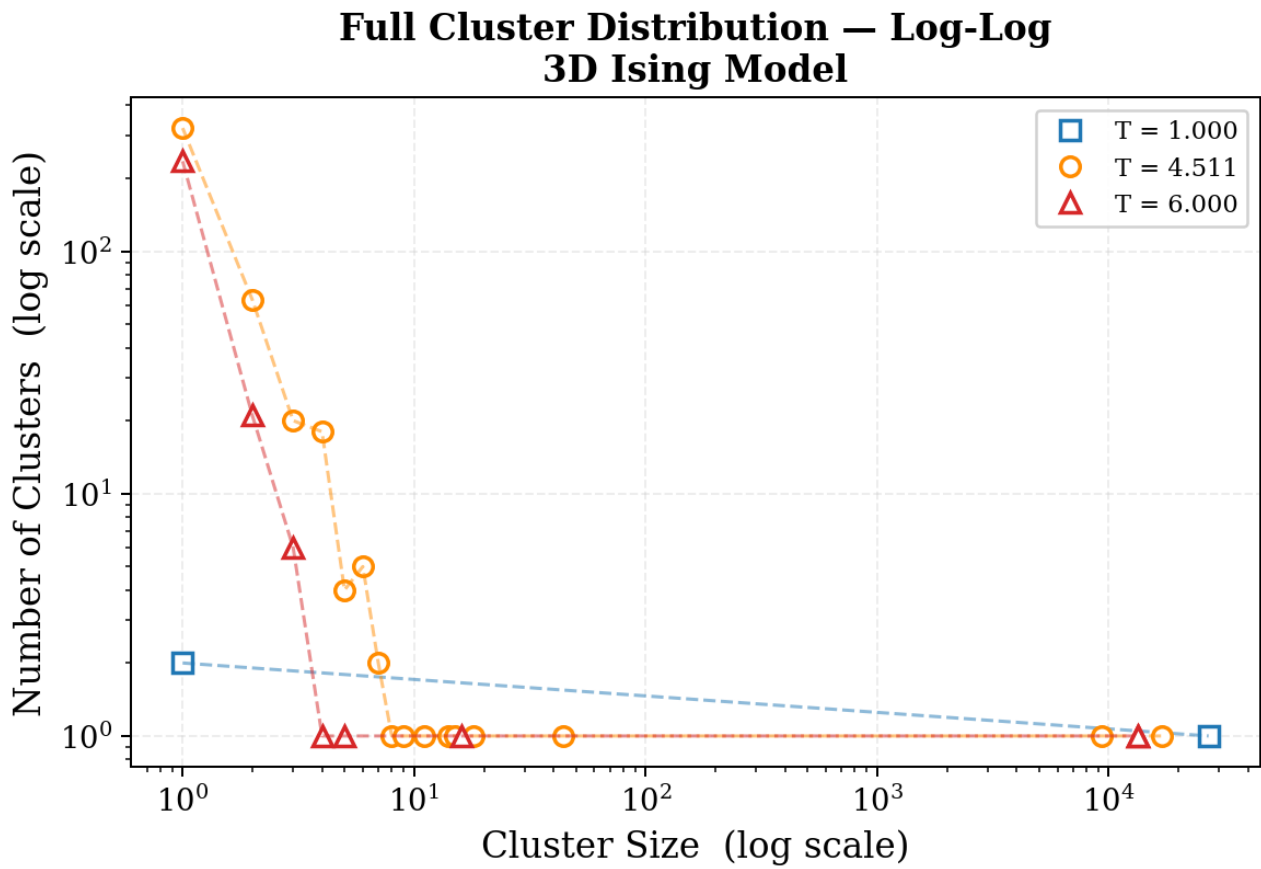
The logarithmic scale makes the variation of the structure factor easier to observe over a wide range of values. Near the critical temperature, the structure factor remains large for small values of k , showing the existence of large correlated regions. At higher values of k , the structure factor decreases because smaller length scales contribute less to the ordering.

Figure 7 — Cluster Statistics vs Temperature



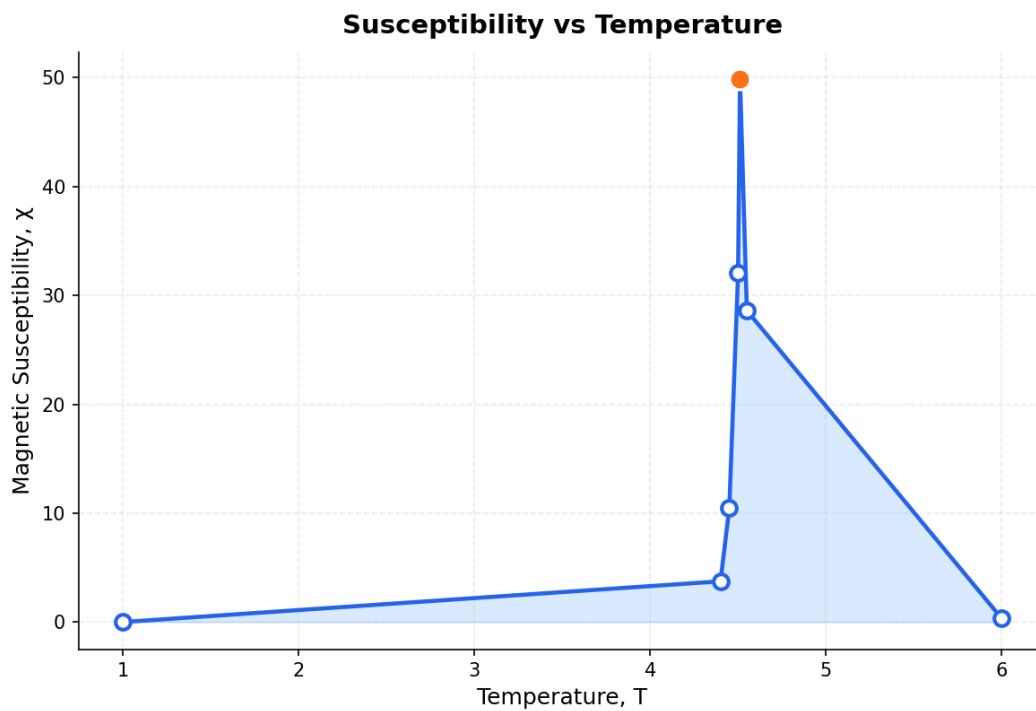
This figure shows the variation of the largest cluster size and the total number of clusters with temperature. At low temperature, most spins belong to one large cluster, giving a very large largest-cluster size. Near the critical temperature, the number of clusters becomes maximum because large and small domains coexist. At high temperature, the largest cluster becomes smaller while the total number of clusters also decreases. This behaviour clearly shows the transition from the ordered phase to the disordered phase.

Figure 8 — Cluster Size Distribution (Log-Log)



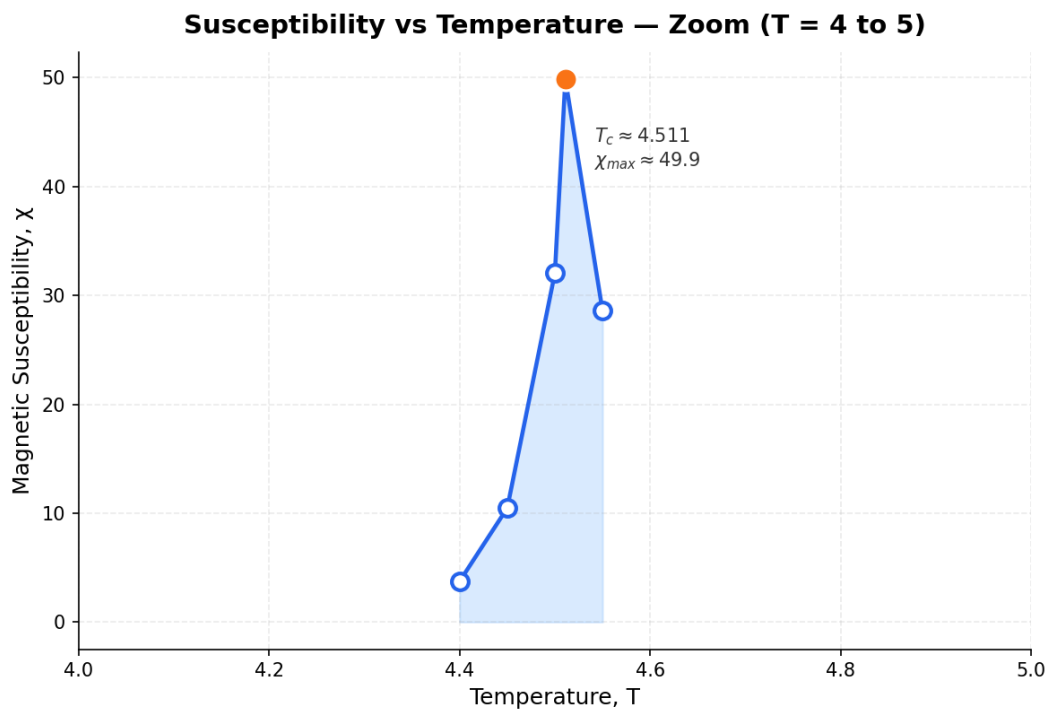
This graph shows the number of clusters of different sizes. At low temperature, one very large cluster dominates the system. Near the critical temperature, clusters of many different sizes appear simultaneously. Above the critical temperature, most clusters become small and the number of large clusters decreases significantly. The broad distribution near the critical temperature indicates strong fluctuations in the system.

Figure 9 — Susceptibility vs Temperature



Magnetic susceptibility was calculated from the fluctuations in magnetization using 50 ensemble measurements at each temperature. The susceptibility is very small at low temperatures, increases rapidly as the temperature approaches the critical region, and reaches a maximum near $T \approx 4.511$. At higher temperatures, the susceptibility decreases as the system enters the disordered phase.

Figure 10 — Susceptibility vs Temperature (Zoomed, $T = 4$ to 5)



This is a zoomed view of the critical region shown in Figure 9, focusing on the temperature range $T = 4$ to 5. The susceptibility rises sharply as the temperature approaches $T_c = 4.511$ and falls just as rapidly beyond it, illustrating the narrow window over which critical fluctuations dominate the system's response.